Chapter 11

Image Sizing and Resolution Independence

DaVinci Resolve is a resolution-independent application. This means that, whatever the resolution of your source media, it can be output at whatever other resolution you like, and just about every size-dependent effect in your project, text, windows of grades, edit and input clip scaling, and other effects will scale appropriately to match the new output resolution.

This also means that you can freely mix clips of any resolution, fitting 4K, HD, and SD clips into the same timeline, with each scaling to fit the project resolution as necessary.

Your project's resolution can be changed at any time, allowing you to work at one resolution, and then output at another resolution. This also makes it easy to output multiple versions of a program at different resolutions, for example, outputting 4K, HD, and SD sized versions of the same timeline.

Additionally, most controls that let you transform clips, either to push into a clip for creative intent, or to pan and scan media of one format to fit better into a different output format, are smart enough to always refer back to the source resolution when combining resizing operations to shrink, then enlarge an image for various reasons as you work in the Cut, Edit, Fusion, and Color pages.

This chapter covers the relationship among the different sizing and transform controls found in DaVinci Resolve, showing how they work together to intelligently manage the sizing of clips and effects as you work.

Contents

About Resolution Independence 283
Timeline Resolution 283
Mixing Clip Resolutions 284
Changing the Timeline Resolution 284
You Can Use Separate Timelines to Output Different Resolutions 284
You Don't Need Separate Timelines to Output Different Resolutions 284
Using High Resolution Media in Lower Resolution Projects 285

Clip Source Resolution 285
Pixel Aspect Ratio (PAR) 285
Clip Resolution 285
The DaVinci Resolve Sizing Pipeline 286
“Super Scale” High Quality Upscaling (Studio Version Only) 286
Fusion Effects and Resolution 287
Image Scaling 289
Edit Sizing in the Cut and Edit pages 293
Image Stabilization 294
Input Sizing on the Color Page 294
Node Sizing on the Color Page 294
Output Sizing on the Color Page 294
Output Blanking 295
Format Resolution on the Delivery Page 295
Rendering Sizing Adjustments and Blanking 296

MEDIA

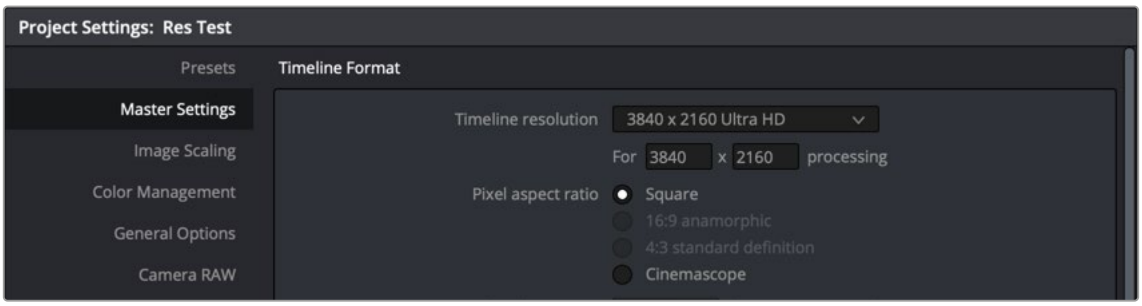
About Resolution Independence

If you only read one paragraph of this chapter, read this: Resolution Independence in DaVinci Resolve means you can add clips to a timeline in any combination of resolutions to fit the project resolution you’ve chosen to work at, and you can later output that timeline to as many other resolutions as necessary in order to create multiple deliverables. When you do so, all effects and transforms will automatically readjust themselves to match the sizing of each new timeline resolution, and most transforms are calculated and processed using the full native resolution of the source media you’ve linked to that clip.

In short, what this means is that you can create multiple deliverables in multiple resolutions by simply changing the timeline resolution or by using a lower resolution setting in the Deliver page compared to the timeline resolution when you create a new job to render out, and every effect will be the right size automatically.

Timeline Resolution

The timeline resolution is one of the most fundamental settings of your project, defining its frame size. It’s found in the Master Settings panel of the Project Settings, where you can choose a predefined resolution from the “Timeline resolution” drop-down menu, or you can type a custom resolution into the X and Y fields below.



The project-wide Timeline Resolution parameters found in the Master Settings panel of the Project Settings window

Mixing Clip Resolutions

Media used in a project does not have to match the timeline resolution. In fact, it's extremely common to mix multiple resolutions within the same timeline. Clips that don't match the current resolution will be automatically resized according to the currently selected Image Scaling setting (described below).

Changing the Timeline Resolution

As mentioned earlier, you can change the timeline resolution whenever you like. When you do so, each Edit page transform, Fusion clip effects output, Color page Power Window, Input and Output Sizing adjustment, tracking path, spatial keyframing value, as well as any other other resolution-dependent Resolve FX effect or transform operation in DaVinci Resolve is automatically and accurately scaled to fit the new resolution.

You Can Use Separate Timelines to Output Different Resolutions

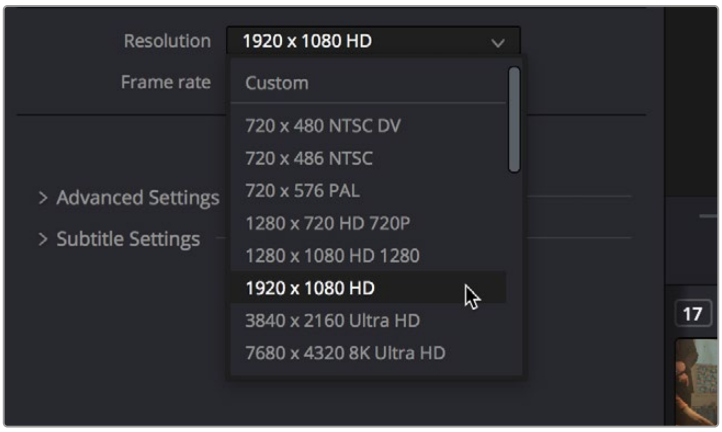
Beginning in DaVinci Resolve 16, you have the option of creating separate timelines with individual Format (including Input Scaling), Monitoring, and Output Sizing settings for situations where you need to set up multiple timelines to create multiple deliverables with different resolutions, pixel aspect ratios, frame rates, monitoring options, or output scaling options than the overall project, including "Mismatched Resolution Files" settings.

For more information, *see Chapter 34, "Creating and Working with Timelines."*

You Don't Need Separate Timelines to Output Different Resolutions

Because of the way DaVinci Resolve works, it's not necessary to create separate timelines when all you need is to output the same timeline at multiple resolutions. Instead, you can focus on mastering a single timeline, which you can output to as many other resolutions as you need.

For example, with only a single timeline in a project set to 4096x2160 (4K DCI) resolution, you can easily output UHD, HD, center-cut SD, and center-cut Instagram sized deliverables in any format you need by simply changing the Resolution drop-down setting in the Deliver page Render Settings before you create a job to render. DaVinci Resolve takes care of the rest.



The Deliver page drop-down menu in the Render Settings panel lets you choose what resolution you want to output the current timeline using

MEDIA

Using High Resolution Media in Lower Resolution Projects

Every set of transform and sizing parameters and settings that resize clips is combined intelligently, so that the full resolution of a clip's source media is always used as the source for any transform. For example, if you're using 8K media within a 1920x1080 project, and you need to enlarge a clip using the Input Sizing palette's Zoom parameter to 200%, the image is scaled relative to the native 8K resolution of the source, and the result is fit into the current timeline resolution. This automatically guarantees the highest quality for any image transform you make so long as you don't zoom in past the native resolution of any given clip.

This also applies to situations where, for example, you shrink a clip in the Edit page using the Edit Sizing controls, only to re-enlarge the same clip in the Color page, using the Input Sizing controls. In this situation, DaVinci Resolve is smart enough to do the math combining the project resolution, the Edit Sizing, and the Input Sizing controls so that a single transform is applied to the native source resolution of that clip, giving you the best quality result.

NOTE: This changes when you apply Fusion effects to any clip, as described later in this chapter.

Clip Source Resolution

Clip resolution in DaVinci Resolve is handled by the combination of Pixel Aspect Ratio and Resolution.

Pixel Aspect Ratio (PAR)

The Timeline Format settings, found in the Master Settings of the Project Settings, let you specify a Pixel Aspect Ratio for the project, in addition to the frame size. This setting defaults to Square Pixel, which is appropriate for high definition projects and most digital media. However, there are also options for 16:9 anamorphic, 4:3 standard definition, or Cinemascope. Which options are available depends on what timeline resolution you've selected.

In addition, each clip has individually adjustable PAR settings in the clip attributes, for situations where you're mixing multiple types of media within a single project. For example, if you're mixing SD clips with non-square pixels and HD clips with square pixels, you can sort out all of the SD clips in the Media Pool and assign them the appropriate NTSC or PAL non-square pixel ratio PAR setting.

For more information, *see Chapter 22, "Modifying Clips and Clip Attributes."*

Clip Resolution

Ordinarily, the resolution of a clip is entirely dependent on the resolution that was selected when that media was shot, or rendered out of a compositing, VFX, or 3D application. Once a piece of media has been created, the native resolution of that media cannot be changed, and to maintain the ideal amount of sharpness for that clip, you need to make sure that whatever transforms you apply to resize a clip zoom into that clip no more than 10-20% over its native resolution (if even that), otherwise the image will visibly soften.

However, DaVinci Resolve provides advanced Super Scale image processing in the Clip Attributes of every video and image clip, that make it possible to resize clips beyond their native resolution while maintaining the perceptible sharpness of a clip that's still within it's native resolution. This is an illusion, but it's a convincing one.

The DaVinci Resolve Sizing Pipeline

This section discusses the various sizing controls that are available in DaVinci Resolve, and how they work together.

"Super Scale" High Quality Upscaling (Studio Version Only)

For instances when you need higher-quality upscaling than the standard Resize Filters allow, you can now enable one of the "Super Scale" options in the Inspector or in the Video panel of the Clip Attributes window for one or more selected clips. Unlike using one of the numerous scaling options in the Edit, Fusion, or Color pages, Super Scale actually increases the source resolution of the clip being processed, which means that clip will have more pixels than it did before and will be more processor-intensive to work with than before, unless you optimize the clip (which bakes in the Super Scale effect into the optimized media) or cache the clip in some way.

Super Scale options in the Video panel of the Clip Attributes

The Super Scale dropdown menu provides the options of 2-3-4x, and 2-3-4x Enhanced, as well as Sharpness and Noise Reduction options to tune the quality of the scaled result. Most of the Super Scale parameters are in fixed Low, Medium, or High increments, however the Enhanced modes let you apply them in variable amounts. Selecting one of these options enables DaVinci Resolve to use advanced algorithms to improve the appearance of image detail when enlarging clips by a significant amount, such as when editing SD archival media into a UHD timeline, or when you find it necessary to enlarge a clip past its native resolution in order to create a closeup.

You may find that, depending on the source media you're working with, setting Sharpness to Medium yields a relatively subtle result that can be hard to notice, but setting Sharpness to high should be immediately more preferable, while also sharpening grain and noise in the image to an undesirable extent at the default settings. However, while raising Noise Reduction will ameliorate this effect, it will also diminish the gains you obtained by raising Sharpness. In these cases, it's worth experimenting with keeping Sharpness at Low or Medium so that Super Scale sharpens all aspects of a clip, but then using the Noise Reduction tools of the Color page (with their additional ability to be fine-tuned) to diminish the unwanted noise.

TIP: Super Scale, while incredibly useful, is a processor-intensive operation, so be aware that turning this on will likely prevent real-time playback. One way to get around this is to create Optimized Media for clips in which you've enabled Super Scale, since Optimized Media "bakes in" the Super Scale effect. Another way to work is to create a string-out of all of the source media you'll need to enlarge at high-quality, turn on Super Scale for all of them, and then render that timeline as individual clips, while turning on the "Render at source resolution" and "Filename uses > Source Name" options.

Fusion Effects and Resolution

All image processing by the Fusion page takes place before effects that are applied by the Edit page, with the sole exception of the Lens Correction effect. When it comes to sizing and image resolution, whether or not the Fusion page affects resolution depends on how you use it.

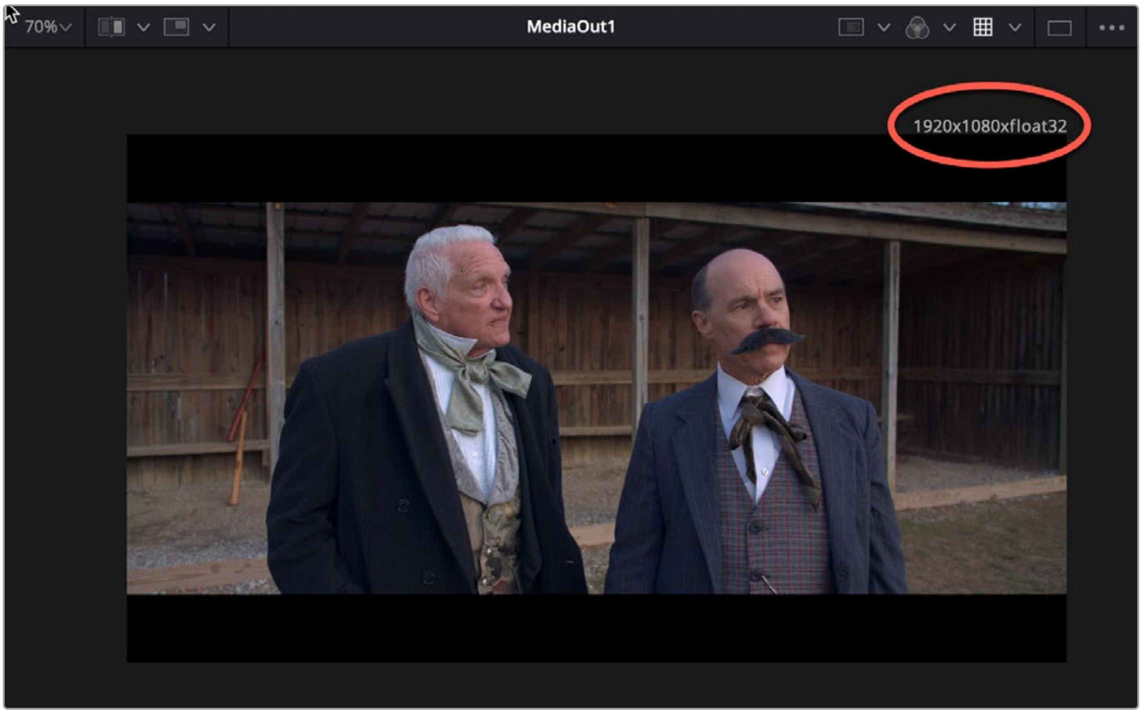
Fusion Effects Inherit the Source Resolution of a Clip

When you open a clip on the Timeline in the Fusion page, the Fusion page is set to the full source resolution of that clip, regardless of the Timeline resolution. This can be seen if you look at the resolution that's listed above the upper right-hand corner of the Viewer. This means that if you don't apply any operations that reduces the image resolution (described later), subsequent sizing adjustments in other pages will refer to the same resolution as the source clip.

The available resolution and bit depth of the currently selected clip is visible above the upper right-hand corner of the Viewer, circled in red

Fusion Clips Inherit the Timeline Resolution

If you combine multiple clips on the Timeline into a Fusion clip, the Fusion page is set to the timeline resolution, regardless of the source resolution of the clip. The image is then output to the Edit page at this timeline resolution, and all subsequent sizing adjustments are performed relative to the timeline resolution, with no reference to the original resolution in the source clip.



The available resolution and bit depth of a clip that's been turned into a Fusion Clip, that's set to the timeline resolution of 1920x1080

Operations in the Fusion Page That Change Resolution

If you don't do anything to change the size of a clip in the Fusion page, then its resolution stays the same and you'll effectively output the source resolution of that clip to the Edit page.

However, if you Merge the image with a second clip attached to the background which has a different resolution, or if you use a Crop or Resize node to increase or decrease the resolution of the image, then the new resolution will be passed to the Cut and Edit pages as the effective source resolution of that clip.

In short, the Fusion page passes whatever resolution is output by the last node of your composition back to the Edit page as the effective resolution of that clip in the DaVinci Resolve image processing pipeline.

Fusion Page Transform Operations Are Resolution Independent

Within the Fusion page, multiple Transform nodes operate in a resolution independent manner relative to the resolution of the source clip. This means that if you shrink an image to 20% with one Transform node, and then enlarge it back up to 100% using a second Transform node, you end up with an image that has all the resolution and sharpness of the input image.

Fusion Page Resize Operations Are Not

Within Fusion there are two kinds of transform effects, the Transform node and the Resize node. Which of these nodes you use has a dramatic impact on resolution independence.

- The Transform node always refers back to the input resolution of the clip (as defined by the Clip Attributes) to enable resolution-independent sizing, such that multiple Transform nodes can scale the image down and up repeatedly within the Fusion page with no unnecessary loss of image resolution.
- The Resize node actually decreases image resolution when you shrink an image or increases image resolution (with filtering) when enlarging. This means that the Resize node will break resolution independence, and the resolution of the image will be fixed at whatever you specify from that point in your composite's node tree forward.

In most situations, you probably want to use the Transform node to maintain resolution independence relative to the source media, unless you specifically want to alter and perhaps reduce image resolution to create a specific special effect which purposefully degrades the image. For example, if you want a clip to be forced to a standard definition resolution in order to make it look like a low-resolution archival clip, the Resize node will accomplish this. Enlarging the result with a Transform node will then perform a filtered enlargement that will look like a real SD clip being enlarged.

Transforms from the Fusion Page to the Edit Page

All transform operations you apply on the Cut, Edit, and Color pages are resolution independent, referring to the original resolution of the source media, so long as you don't use the Fusion page. For example, if you shrink an image to 20% in the Edit page (using Edit sizing controls) and then enlarge it in the Color page back to 100% (using Input sizing controls), you end up with an image that has all the resolution and sharpness of the original media, because the final resolution is drawn from the original source media.

However, once you use the Fusion page to do anything to a clip, from adding a small effect to creating a complex composition, the resolution-independent relationship of the Edit and Color pages to the source media is broken, and whatever resolution is output from your Fusion composition is the new effective resolution of the clip that appears in the Timeline. This means if you shrink an image to 20% in the Fusion page (using a Transform node) and then enlarge it in the Color page by 150%, you end up with an image that isn't as sharp as the original because the downconverted image in the Fusion page is effectively the new source resolution of that clip.

Image Scaling

DaVinci Resolve has a dedicated mechanism for automatically managing the size of clips with resolutions that don't match the timeline resolution, and it's separate from the Zoom transform controls that are available for making creative adjustments to clips. This is called Image Scaling, and it's customizable in a few different areas.

Resize Filter Project Setting

The Resize Filter setting lets you choose the filter method that's used to interpolate image pixels when resizing clips:

Smoother: May provide a more pleasing result for projects using clips that must be scaled down to standard definition as this filter exhibits fewer sharp edges at SD resolutions.

Bicubic: While the sharper and smoother options are slightly higher quality, bicubic is still an exceptionally good resizing filter and is less processor-intensive than either of those options.

Blinear: A lower quality setting that is less processor-intensive. Useful for previewing your work on a low-performance computer before rendering when you can switch to one of the higher quality options.

Sharper: Usually provides the best quality for most projects, using an optical quality processing technique that's unique to DaVinci Resolve.

Custom: This setting lets you take control of the exact algorithm used in all resizing operations. The custom Resize Filter options available are: Bessel, Box, Catmul-Rom, Cubic, Gaussian, Lanczos, Mitchell, Nearest Neighbor, Quadratic, and Sinc. In practice, the difference between these methods can be quite subjective. However, if you need to match a specific resizing method used from another application, you can do it here. For everyday use, the normal resizing filters in DaVinci Resolve should be sufficient.

Override Input scaling: Checking this box lets you choose an Input Sizing preset to apply to the project.

Override Output scaling: Checking this box lets you choose an Output Sizing preset to apply to the project.

Anti-alias edges: A second group of settings lets you choose how to handle edge anti-aliasing for source blanking.

- **Auto:** Adds anti-aliasing when any of the Sizing controls are used to transform the image. Otherwise, anti-aliasing is disabled.
- **On:** Forces anti-aliasing on at all times.
- **Off:** Disables anti-aliasing. It might be necessary to turn anti-aliasing off if you notice black blurring at the edges of blanking being applied to an image.

Deinterlace quality: (only available in Studio version) A fourth group of settings lets you choose the quality/processing time tradeoff when deinterlacing Media Pool clips using the Enable Deinterlacing checkbox in the Clip Attributes window. There are two settings:

- **Normal:** A high-quality deinterlacing method that is suitable for most clips. For many clips, Normal is indistinguishable from High. Normal is always used automatically during playback in DaVinci Resolve.
- **High:** A more processor-intensive method that can sometimes yield better results, depending on the footage, at the expense of slower rendering times.
- **DaVinci Neural Engine:** This option uses the advanced machine learning algorithms of the DaVinci Neural Engine to analyze motion between the fields of interlaced material and reconstructs them into a single frame. This option is very computationally intensive, but ideally will deliver an even more aesthetically pleasing result than the "high" setting.

Input Scaling Project Setting

If the native resolution of an imported clip doesn't match the timeline resolution, then the currently selected Input Scaling Preset in the Image Scaling panel of the Project Settings dictates how mismatched clips will be handled project-wide. The default setting is "Scale entire image to fit," which shrinks or enlarges the image to fit the current dimensions of the frame without cropping any part of the image, adding letterboxing or pillarboxing as necessary to fill the unused portion of the frame depending on whether the horizontal or vertical dimension of the image hits the edge of the frame first.

The Mismatched resolution files option let you choose how clips that don't match the current project resolution are handled. The illustrated examples below show an SD clip being fit into an HD project using each of the different options.

Center crop with no resizing

Clips of differing resolution are not scaled at all. Clips that are smaller than the current frame size are surrounded by blanking, and clips that are larger than the current frame size are cropped. Keep in mind that this is a good setting to use if you're importing a timeline from another NLE in which clip resolution adjustments are imported as scaling adjustments. Choosing "Center Crop with no resizing" prevents DaVinci Resolve from "double scaling" clips in imported timelines.

Center crop with no resizing

Scale full frame with crop

Clips of differing resolution are scaled so that the clip fills the frame with no blanking. Excess pixels are cropped. This is a good setting when you want clips that don't match the project resolution to automatically fill the frame, with no letterboxing or pillarboxing.

Scale full frame with crop

Stretch frame to all corners

Useful for projects using anamorphic media. Clips of differing resolutions are squished or stretched to match the frame size in all dimensions. This way, anamorphic media can be stretched to match full raster or full raster media can be squished to fit into an anamorphic frame. An added benefit of this setting is that it makes it easy to mix anamorphic and non-anamorphic clips in the same project.

Stretch frame to all corners

Scale entire image to fit

The default setting. Clips of differing resolution are scaled so that each clip fills the frame without cropping. The dimension that falls short has blanking inserted (letterboxing or pillarboxing). This is a good setting when you want clips that don't match the project resolution to automatically fit into the frame without being cropped in any way, and you're fine with letterboxing or pillarboxing as a result. However, if you've imported a timeline from another NLE and there are clips that are twice as big as they should be, it's because this setting is on by default, and your imported timeline has imported scaling settings used to resize clips that didn't match the timeline resolution. If this happens, switch to "Center crop with no resizing" instead, and that will fix the problem.

Scale entire image to fit

MEDIA

Output Image Scaling Project Settings

Another group of settings found in the Image Scaling panel of the Project Settings lets you optionally choose a different resolution to be output, either via the Deliver page, or via your video output interface for monitoring or outputting to tape.

In particular, if you set the "Resolution" in the Render Settings panel of the Deliver page to something other than the timeline resolution, these settings are used to make the change. This is useful in situations where you're mastering a high resolution 4K project, but you want to monitor using an HD display, and you plan on eventually outputting HD resolution deliverables in addition to the 4K deliverables for which you want to use different Scaling and/or Resize Filter settings that work better at the lower resolution.

Match timeline settings: This checkbox is turned on by default so that these settings mirror the Image Scaling and Input Scaling settings described above. Turning this checkbox off lets you choose different settings to be used when monitoring, outputting to tape, or rendering, using the other settings below.

Output resolution: Lets you choose an alternate resolution for monitoring and delivery. You can also set this from the "Resolution" drop-down menu of the Video panel in the Render Settings of the Deliver page.

For "X x Y" processing: Lets you specify a different custom alternate resolution.

Pixel aspect ratio: Lets you specify an alternate pixel aspect ratio to match the alternate timeline format.

Mismatched resolution files: Lets you choose an alternate way of handling mismatched resolution files that works better for the alternate resolution you've chosen. These options work similarly to those of the "Input Scaling" group. For example, for an HD or UHD resolution project you may have the Image Input Scaling set to "Scale Full Frame With Crop" so that all Standard Definition resolution files are center-cut to eliminate blanking. However, if you're using Output Image Scaling to create a Standard Definition deliverable, you may want to set the Output Image Scaling > Mismatched resolution files setting to “Scale entire image to fit” in order to letterbox all HD or UHD resolution clips, while preserving the original aspect ratio of the SD clips.

Super Scale: Sets a very processor-intensive and high quality upscaling algorithm that actually creates new pixels for the resized image. The possible values are None, 2-3-4x, 2-3-4x Enhanced, and Auto.

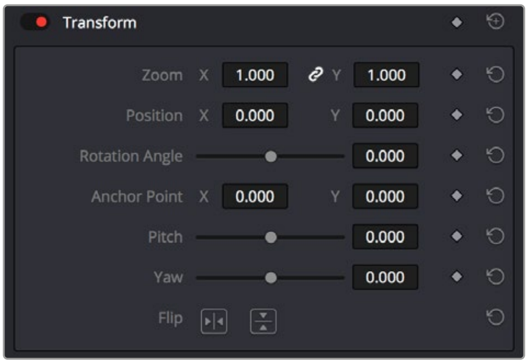
Clip-Specific Scaling Settings

There’s an additional set of Scaling and Resize Filter settings, available in the Video Inspector for selected clips, that provide the same options as those found in the Project Settings window, except that they let you choose settings that will be specific to a particular clip. These are valuable for situations where the project-wide scaling setting is working for most clips, but you have a handful of specific clips that would benefit from individual settings.

Edit Sizing in the Cut and Edit pages

The Video Inspector contains a set of Transform parameters with which you can alter clips in the Timeline. These parameters operate independently of the Input Sizing controls found in the Color page. Separate Edit sizing controls serve a number of different functions:

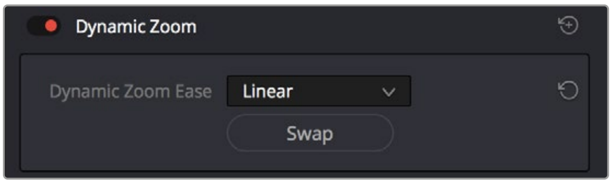
- They’re convenient for editors and are easily animated for creating motion graphics effects right on the Cut and Edit page timelines. They also keep editor transform adjustments separate from colorist transform adjustments, for a clear division of labor and responsibility.
- Edit sizing parameters also store incoming transform data from imported AAF and XML projects that come from other applications, so that imported transforms are kept separate from adjustments made by colorists and finishing artists.



The Transform parameters in the Inspector of the Edit page

If, when importing an AAF or XML project file, you turned on the “Use sizing information” checkbox, then every clip that had position, scale, rotation, or crop settings applied in the originating NLE will have those adjustments applied to these transform parameters, which is convenient for keeping imported transform settings separate from other DaVinci Resolve-native transform settings.

Additionally, a set of Dynamic Zoom parameters also exists in the Video Inspector, which let you make quickly animated transforms using graphical controls that correspond to the start and end states of an animated transform. However, these transforms are lumped in with the other Edit page Transform parameters in terms of the order of sizing operations occurring throughout DaVinci Resolve.



The Dynamic Zoom settings in the Inspector of the Video Inspector

The transform that's made via the Edit Sizing controls refers back to either the source resolution of each clip, or the resolution output by the Fusion page if it's in use.

Image Stabilization

DaVinci Resolve provides Image Stabilization controls in the Cut, Edit, and Color pages that all control the same transform operation that happens between Edit sizing and Input Sizing in the image processing pipeline. The transform that's made via the Image Stabilization controls refers all the way back to either the source resolution of each clip, or the resolution output by the Fusion page if it's in use.

Input Sizing on the Color Page

The Sizing palette on the Color page has another dedicated set of keyframable transform parameters that work with the various DaVinci control panels to let the colorist apply pan and scan adjustments while working through a project. These parameters work independently of the Edit page Transform parameters, allowing you to keep imported transform settings separate from other transform settings that you apply. However, for convenience the Edit sizing controls are available in the Color page as well.

The transform that's made via the Input Sizing controls refers all the way back to either the source resolution of each clip, or the resolution output by the Fusion page if it's in use.

Node Sizing on the Color Page

Using Node Sizing, you can apply individual sizing adjustments to clips on a per-node basis within the Color page, which is similar in principal to using Transform nodes in the Fusion page. All Node Sizing adjustments within a grade are cumulative, and any keyframing done to Node Sizing parameters is stored in that node's Node Format keyframe track in the Keyframe Editor. Two good examples of Node Sizing include realigning color channels individually in conjunction with the Splitter/Combiner nodes or duplicating windowed regions of an image by moving them around the frame. Subsequent Node Sizing operations do not refer back to the source resolution of a clip, so using multiple Node Sizing operations to reduce and enlarge an image will reduce image resolution and sharpness.

Output Sizing on the Color Page

Output sizing is an additional transform that is applied after Edit sizing, Fusion sizing, Input sizing, and Node sizing. It's an overall adjustment that affects every clip at once, which is suitable for making last-minute format alterations that you want to affect the entire program. Technically, Output Sizing includes the Blanking controls, but those are important enough to discuss separately. Output Sizing also does not refer back to the source resolution of clips, so if you use Edit or Input Sizing to shrink a clip, and Output Sizing to enlarge it again, the final result will be somewhat softened as you're enlarging the lower resolution image output by Input Sizing.

Output Blanking

Output blanking is not a sizing operation, but it's often related and so worth mentioning here. Blanking is an adjustment you can use to add black areas to the top, bottom, left, or right of an image, in order to add "letterboxing" (black bars at the top and bottom of the image) or "pillarboxing" (black bars at the left and right of the image) that lets you fill in the unused parts of an image frame that's either shorter or thinner than the current output resolution.

Once all transforms, compositing operations, and color corrections have been applied by the DaVinci Resolve image processing pipeline, the very last operation to be performed is Output blanking, if it's enabled. This guarantees that overlapping images, grading, and other adjustments are properly "blacked out" no matter what you're doing to the program.

Output Blanking controls are found in the Timeline menu (as a series of aspect ratios) as well as in the Output Sizing parameters of the Color page Sizing palette (via Top, Right, Bottom, and Left controls).

TIP: Text and graphics superimposed via the Data Burn-In window, if enabled, are the only effects that will appear in front of picture areas affected by blanking. This lets you add timecode and other information over letterboxed areas that you don't want to obscure the picture.

Format Resolution on the Delivery Page

By default, the Format Resolution setting in the Render Settings of the Deliver page matches the timeline resolution when "Match timeline settings" is enabled in the Output Scaling Preset in the Image Scaling panel of the Project Settings.

Choosing a new resolution from the "Set Resolution to" drop-down menu lets you override the current Format Resolution setting before rendering. Using this control, you can queue up multiple jobs, each set to a different resolution, to output multiple formats during a single render session.

For more information on rendering and setting up jobs for the Render Queue, *see Chapter 186, "Using the Deliver Page."*

Rendering Sizing Adjustments and Blanking

When rendering your final output, you have the option of choosing whether or not to “bake in” the sizing operations that have been performed. For example, you may have set up a whole set of specific sizing adjustments for the clips in a program, but then you’re requested to render the project and its media as individual clips for round trip re-delivery to the editor for further work. In this case, you can choose to either render the sizing into the final media, or not.

Whether or not sizing is rendered into your final media depends on the “Disable edit and input sizing” checkbox in the Advanced Settings options of the Render Settings panel. You can disable sizing and blanking either when rendering the current timeline as a single clip, or when rendering individual clips.

If “Disable sizing and blanking output” is turned off: Output Blanking, Cut and Edit page sizing adjustments, Color page Input and Output Sizing adjustments, and Image Stabilization are rendered into the final rendered media using the optical-quality sizing algorithms available to DaVinci Resolve. This is best if your sizing adjustments are approved and final, and you want to “bake” sizing adjustments into the final media you’re delivering.

If “Disable sizing and blanking output” is turned on: Output Blanking, Cut and Edit page sizing adjustments, Color page Input and Output Sizing adjustments, and Image Stabilization are not rendered, and each clip will be rendered either at the source resolution if “Render at source resolution” is enabled in individual clips mode, or to the currently specified resolution of the timeline or project. However, the sizing adjustments you’ve made will be exported as part of the XML or AAF file that you’re exporting. This is best for workflows where the editor wants to continue adjusting sizing after you’ve handed off the graded project relative to the original size of the clips.

Keep in mind that if you want to render Input Sizing adjustments into the media you’re outputting, the “Force sizing to highest quality” checkbox guarantees that DaVinci Resolve will use the highest-quality sizing setting, even if you’ve temporarily chosen a faster-processing option for a slower computer.

NOTE: “Disable sizing and blanking output” does not disable any transform operations that happen within the Fusion page. Those will continue to be applied to the final output.